67. $(5.89d + 2.75d) + 1.25d$

68. $17\,(0.25)\,(4)$

69. $36\,(0.2)\,(5)$

70. $[2.48\,(12)]\,(0.5)$

71. $[9.731\,(4)]\,(0.75)$

72. $7\,(4a)$

73. $9\,(8w)$

74. $-15\,(5m)$

75. $-23\,(2n)$

76. $12\left(\frac{5}{6}p\right)$

77. $20\left(\frac{3}{5}q\right)$

78. $14x + 19y + 25x + 3y$

79. $15u + 11v + 27u + 19v$

80. $43m + (-12n) + (-16m) + (-9n)$

81. $-22p + 17q + (-35p) + (-27q)$

82. $\frac{3}{8}g + \frac{1}{12}h + \frac{7}{8}g + \frac{5}{12}h$

83. $\frac{5}{6}a + \frac{3}{10}b + \frac{1}{6}a + \frac{9}{10}b$

84. $6.8p + 9.14q + (-4.37p) + (-0.88q)$

85. $9.6m + 7.22n + (-2.19m) + (-0.65n)$

Everyday Math

86. **Stamps** Allie and Loren need to buy stamps. Allie needs four $0.49 stamps and nine $0.02 stamps. Loren needs eight $0.49 stamps and three $0.02 stamps.

ⓐ How much will Allie's stamps cost?

ⓑ How much will Loren's stamps cost?

ⓒ What is the total cost of the girls' stamps?

ⓓ How many $0.49 stamps do the girls need altogether? How much will they cost?

ⓔ How many $0.02 stamps do the girls need altogether? How much will they cost?

87. **Counting Cash** Grant is totaling up the cash from a fundraising dinner. In one envelope, he has twenty-three $5 bills, eighteen $10 bills, and thirty-four $20 bills. In another envelope, he has fourteen $5 bills, nine $10 bills, and twenty-seven $20 bills.

ⓐ How much money is in the first envelope?

ⓑ How much money is in the second envelope?

ⓒ What is the total value of all the cash?

ⓓ What is the value of all the $5 bills?

ⓔ What is the value of all $10 bills?

ⓕ What is the value of all $20 bills?

Writing Exercises

88. In your own words, state the Commutative Property of Addition and explain why it is useful.

89. In your own words, state the Associative Property of Multiplication and explain why it is useful.

Self Check

ⓐ After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
use the Commutative and Associative Properties.			
evaluate expressions using the Commutative and Associative Properties.			
simplify expressions using the Commutative and Associative Properties.			

ⓑ After reviewing this checklist, what will you do to become confident for all objectives?

7.3 Distributive Property

Learning Objectives

By the end of this section, you will be able to:

- Simplify expressions using the distributive property
- Evaluate expressions using the distributive property

☑ **BE PREPARED** 7.7 Before you get started, take this readiness quiz.

Multiply: $3\,(0.25)$.

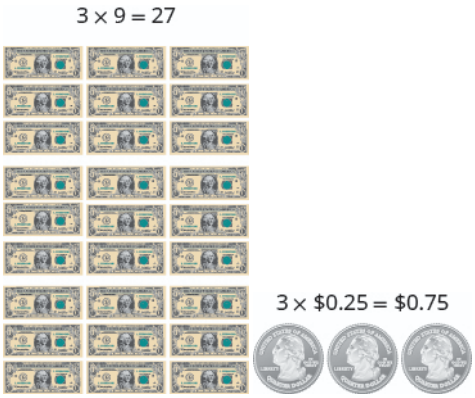
If you missed this problem, review Example 5.15

☑ **BE PREPARED** 7.8 Simplify: $10-(-2)(3)$.
If you missed this problem, review Example 3.51

☑ **BE PREPARED** 7.9 Combine like terms: $9y+17+3y-2$.
If you missed this problem, review Example 2.22.

Simplify Expressions Using the Distributive Property

Suppose three friends are going to the movies. They each need $\$9.25$; that is, 9 dollars and 1 quarter. How much money do they need all together? You can think about the dollars separately from the quarters.



They need 3 times $\$9$, so $\$27$, and 3 times 1 quarter, so 75 cents. In total, they need $\$27.75$.

If you think about doing the math in this way, you are using the Distributive Property.

Distributive Property

If a, b, c are real numbers, then

$$a\,(b+c)=ab+ac$$

Back to our friends at the movies, we could show the math steps we take to find the total amount of money they need like this:

$$\begin{aligned} & 3(9.25) \\ 3(9 & + 0.25) \\ 3(9) & + 3(0.25) \\ 27 & + 0.75 \\ & 27.75 \end{aligned}$$

In algebra, we use the Distributive Property to remove parentheses as we simplify expressions. For example, if we are asked to simplify the expression $3(x+4)$, the order of operations says to work in the parentheses first. But we cannot add x and 4, since they are not like terms. So we use the Distributive Property, as shown in Example 7.17.

EXAMPLE 7.17

Simplify: $3\,(x+4)$.

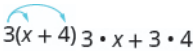
Solution

	$3(x+4)$
Distribute.	$3 \cdot x + 3 \cdot 4$
Multiply.	$3x + 12$

TRY IT 7.33 Simplify: $4(x+2)$.

TRY IT 7.34 Simplify: $6(x+7)$.

Some students find it helpful to draw in arrows to remind them how to use the Distributive Property. Then the first step in Example 7.17 would look like this:



EXAMPLE 7.18

Simplify: $6(5y+1)$.

Solution

	$6(5y+1)$
Distribute.	$6 \cdot 5y + 6 \cdot 1$
Multiply.	$30y + 6$

TRY IT 7.35 Simplify: $9(3y+8)$.

TRY IT 7.36 Simplify: $5(5w+9)$.

The distributive property can be used to simplify expressions that look slightly different from $a(b+c)$. Here are two other forms.

Distributive Property

If a, b, c are real numbers, then

$$a(b+c) = ab + ac$$

Other forms

$$a(b-c) = ab - ac$$
$$(b+c)a = ba + ca$$

EXAMPLE 7.19

Simplify: $2\,(x-3)$.

Solution

$2(x-3)$

Distribute.	$2 \cdot x - 2 \cdot 3$
Multiply.	$2x - 6$

TRY IT 7.37 Simplify: $7\,(x-6)$.

TRY IT 7.38 Simplify: $8\,(x-5)$.

Do you remember how to multiply a fraction by a whole number? We'll need to do that in the next two examples.

EXAMPLE 7.20

Simplify: $\frac{3}{4}(n+12)$.

Solution

$\frac{3}{4}(n+12)$

Distribute.	$\frac{3}{4} \cdot n + \frac{3}{4} \cdot 12$
Simplify.	$\frac{3}{4}n + 9$

TRY IT 7.39 Simplify: $\frac{2}{5}(p+10)$.

TRY IT 7.40 Simplify: $\frac{3}{7}(u+21)$.

EXAMPLE 7.21

Simplify: $8\left(\frac{3}{8}x + \frac{1}{4}\right)$.

Solution

$8\left(\frac{3}{8}x+\frac{1}{4}\right)$

Distribute.	$8\cdot\frac{3}{8}x+8\cdot\frac{1}{4}$
Multiply.	$3x+2$

TRY IT 7.41 Simplify: $6\left(\frac{5}{6}y+\frac{1}{2}\right)$.

TRY IT 7.42 Simplify: $12\left(\frac{1}{3}n+\frac{3}{4}\right)$.

Using the Distributive Property as shown in the next example will be very useful when we solve money applications later.

EXAMPLE 7.22

Simplify: $100\,(0.3+0.25q)$.

Solution

$100(0.3+0.25q)$

Distribute.	$100(0.3)+100(0.25q)$
Multiply.	$30+25q$

TRY IT 7.43 Simplify: $100\,(0.7+0.15p)$.

TRY IT 7.44 Simplify: $100\,(0.04+0.35d)$.

In the next example we'll multiply by a variable. We'll need to do this in a later chapter.

EXAMPLE 7.23

Simplify: $m(n-4)$.

Solution

$m(n-4)$

Distribute.	$m\cdot n-m\cdot 4$
Multiply.	$mn-4m$

Notice that we wrote $m\cdot 4$ as $4m$. We can do this because of the Commutative Property of Multiplication. When a term is

the product of a number and a variable, we write the number first.

TRY IT 7.45 Simplify: $r\,(s-2)$.

TRY IT 7.46 Simplify: $y\,(z-8)$.

The next example will use the 'backwards' form of the Distributive Property, $(b+c)a = ba + ca$.

EXAMPLE 7.24

Simplify: $(x+8)\,p$.

Solution

	$(x+8)p$
Distribute.	$px+8p$

TRY IT 7.47 Simplify: $(x+2)\,p$.

TRY IT 7.48 Simplify: $(y+4)\,q$.

When you distribute a negative number, you need to be extra careful to get the signs correct.

EXAMPLE 7.25

Simplify: $-2(4y+1)$.

Solution

	$-2(4y+1)$
Distribute.	$-2\cdot 4y+(-2)\cdot 1$
Simplify.	$-8y-2$

TRY IT 7.49 Simplify: $-3(6m+5)$.

TRY IT 7.50 Simplify: $-6(8n+11)$.

EXAMPLE 7.26

Simplify: $-11(4-3a)$.

Solution

	$-11(4-3a)$
Distribute.	$-11\cdot 4-(-11)\cdot 3a$
Multiply.	$-44-(-33a)$
Simplify.	$-44+33a$

You could also write the result as $33a-44$. Do you know why?

TRY IT 7.51 Simplify: $-5(2-3a)$.

TRY IT 7.52 Simplify: $-7(8-15y)$.

In the next example, we will show how to use the Distributive Property to find the opposite of an expression. Remember, $-a=-1\cdot a$.

EXAMPLE 7.27

Simplify: $-(y+5)$.

Solution

	$-(y+5)$
Multiplying by –1 results in the opposite.	$-1(y+5)$
Distribute.	$-1\cdot y+(-1)\cdot 5$
Simplify.	$-y+(-5)$
Simplify.	$-y-5$

TRY IT 7.53 Simplify: $-(z-11)$.

TRY IT 7.54 Simplify: $-(x-4)$.

Sometimes we need to use the Distributive Property as part of the order of operations. Start by looking at the parentheses. If the expression inside the parentheses cannot be simplified, the next step would be multiply using the distributive property, which removes the parentheses. The next two examples will illustrate this.

EXAMPLE 7.28

Simplify: $8-2\,(x+3)$.

Solution

	$8-2(x+3)$
Distribute.	$8-2\cdot x-2\cdot 3$
Multiply.	$8-2x-6$
Combine like terms.	$-2x+2$

TRY IT 7.55 Simplify: $9-3\,(x+2)$.

TRY IT 7.56 Simplify: $7x-5\,(x+4)$.

EXAMPLE 7.29

Simplify: $4\,(x-8)-(x+3)$.

Solution

	$4(x-8)-(x+3)$
Distribute.	$4x-32-x-3$
Combine like terms.	$3x-35$

TRY IT 7.57 Simplify: $6\,(x-9)-(x+12)$.

TRY IT 7.58 Simplify: $8\,(x-1)-(x+5)$.

Evaluate Expressions Using the Distributive Property

Some students need to be convinced that the Distributive Property always works.

In the examples below, we will practice evaluating some of the expressions from previous examples; in part ⓐ, we will evaluate the form with parentheses, and in part ⓑ we will evaluate the form we got after distributing. If we evaluate both expressions correctly, this will show that they are indeed equal.

EXAMPLE 7.30

When $y=10$ evaluate: ⓐ $6(5y+1)$ ⓑ $6\cdot 5y+6\cdot 1$.

Solution

ⓐ

	$6(5y+1)$
Substitute 10 for y.	$6(5\cdot 10+1)$

Simplify in the parentheses.	$6(51)$
Multiply.	306

ⓑ

	$6 \cdot 5y + 6 \cdot 1$
Substitute 10 for y.	$6 \cdot 5 \cdot 10 + 6 \cdot 1$
Simplify.	$300 + 6$
Add.	306

Notice, the answers are the same. When $y = 10$,

$$6(5y + 1) = 6 \cdot 5y + 6 \cdot 1.$$

Try it yourself for a different value of y.

TRY IT 7.59 Evaluate when $w = 3$: ⓐ $5(5w + 9)$ ⓑ $5 \cdot 5w + 5 \cdot 9$.

TRY IT 7.60 Evaluate when $y = 2$: ⓐ $9(3y + 8)$ ⓑ $9 \cdot 3y + 9 \cdot 8$.

EXAMPLE 7.31

When $y = 3$, evaluate ⓐ $-2(4y + 1)$ ⓑ $-2 \cdot 4y + (-2) \cdot 1$.

Solution

ⓐ

	$-2(4y + 1)$
Substitute 3 for y.	$-2(4 \cdot 3 + 1)$
Simplify in the parentheses.	$-2(13)$
Multiply.	-26

ⓑ

	$-2 \cdot 4y + (-2) \cdot 1$
Substitute 3 for y.	$-2 \cdot 4 \cdot 3 + (-2) \cdot 1$
Multiply.	$-24 - 2$

Subtract.	-26
The answers are the same. When $y = 3$,	$-2(4y + 1) = -8y - 2$

TRY IT 7.61 Evaluate when $n = -2$: (a) $-6(8n + 11)$ (b) $-6 \cdot 8n + (-6) \cdot 11$.

TRY IT 7.62 Evaluate when $m = -1$: (a) $-3(6m + 5)$ (b) $-3 \cdot 6m + (-3) \cdot 5$.

EXAMPLE 7.32

When $y = 35$ evaluate (a) $-(y + 5)$ and (b) $-y - 5$ to show that $-(y + 5) = -y - 5$.

Solution

(a)

	$-(y + 5)$
Substitute 35 for y.	$-(35 + 5)$
Add in the parentheses.	$-(40)$
Simplify.	-40

(b)

	$-y - 5$
Substitute 35 for y.	$-35 - 5$
Simplify.	-40
The answers are the same when $y = 35$, demonstrating that	$-(y + 5) = -y - 5$

TRY IT 7.63 Evaluate when $x = 36$: (a) $-(x - 4)$ (b) $-x + 4$ to show that $-(x - 4) = -x + 4$.

TRY IT 7.64 Evaluate when $z = 55$: (a) $-(z - 10)$ (b) $-z + 10$ to show that $-(z - 10) = -z + 10$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Model Distribution (https://www.openstax.org/l/24ModelDist)

The Distributive Property (https://www.openstax.org/l/24DistProp)

SECTION 7.3 EXERCISES

Practice Makes Perfect

Simplify Expressions Using the Distributive Property

In the following exercises, simplify using the distributive property.

90. $4(x+8)$

91. $3(a+9)$

92. $8(4y+9)$

93. $9(3w+7)$

94. $6(c-13)$

95. $7(y-13)$

96. $7(3p-8)$

97. $5(7u-4)$

98. $\frac{1}{2}(n+8)$

99. $\frac{1}{3}(u+9)$

100. $\frac{1}{4}(3q+12)$

101. $\frac{1}{5}(4m+20)$

102. $9\left(\frac{5}{9}y-\frac{1}{3}\right)$

103. $10\left(\frac{3}{10}x-\frac{2}{5}\right)$

104. $12\left(\frac{1}{4}+\frac{2}{3}r\right)$

105. $12\left(\frac{1}{6}+\frac{3}{4}s\right)$

106. $r(s-18)$

107. $u(v-10)$

108. $(y+4)p$

109. $(a+7)x$

110. $-2(y+13)$

111. $-3(a+11)$

112. $-7(4p+1)$

113. $-9(9a+4)$

114. $-3(x-6)$

115. $-4(q-7)$

116. $-9(3a-7)$

117. $-6(7x-8)$

118. $-(r+7)$

119. $-(q+11)$

120. $-(3x-7)$

121. $-(5p-4)$

122. $5+9(n-6)$

123. $12+8(u-1)$

124. $16-3(y+8)$

125. $18-4(x+2)$

126. $4-11(3c-2)$

127. $9-6(7n-5)$

128. $22-(a+3)$

129. $8-(r-7)$

130. $-12-(u+10)$

131. $-4-(c-10)$

132. $(5m-3)-(m+7)$

133. $(4y-1)-(y-2)$

134. $5(2n+9)+12(n-3)$

135. $9(5u+8)+2(u-6)$

136. $9(8x-3)-(-2)$

137. $4(6x-1)-(-8)$

138. $14(c-1)-8(c-6)$

139. $11(n-7)-5(n-1)$

140. $6(7y+8)-(30y-15)$

141. $7(3n+9)-(4n-13)$

Evaluate Expressions Using the Distributive Property

In the following exercises, evaluate both expressions for the given value.

142. If $v=-2$, evaluate
(a) $6(4v+7)$
(b) $6\cdot 4v+6\cdot 7$

143. If $u=-1$, evaluate
(a) $8(5u+12)$
(b) $8\cdot 5u+8\cdot 12$

144. If $n=\frac{2}{3}$, evaluate
(a) $3\left(n+\frac{5}{6}\right)$
(b) $3\cdot n+3\cdot\frac{5}{6}$

145. If $y=\frac{3}{4}$, evaluate
(a) $4\left(y+\frac{3}{8}\right)$
(b) $4\cdot y+4\cdot\frac{3}{8}$

146. If $y=\frac{7}{12}$, evaluate
(a) $-3(4y+15)$
(b) $-3\cdot 4y+(-3)\cdot 15$

147. If $p=\frac{23}{30}$, evaluate
(a) $-6(5p+11)$
(b) $-6\cdot 5p+(-6)\cdot 11$

148. If $m=0.4$, evaluate
(a) $-10(3m-0.9)$
(b) $-10\cdot 3m-(-10)(0.9)$

149. If $n=0.75$, evaluate
(a) $-100(5n+1.5)$
(b) $-100\cdot 5n+(-100)(1.5)$

150. If $y=-25$, evaluate
(a) $-(y-25)$
(b) $-y+25$

151. If $w=-80$, evaluate
(a) $-(w-80)$
(b) $-w+80$

152. If $p=0.19$, evaluate
(a) $-(p+0.72)$
(b) $-p-0.72$

153. If $q=0.55$, evaluate
(a) $-(q+0.48)$
(b) $-q-0.48$